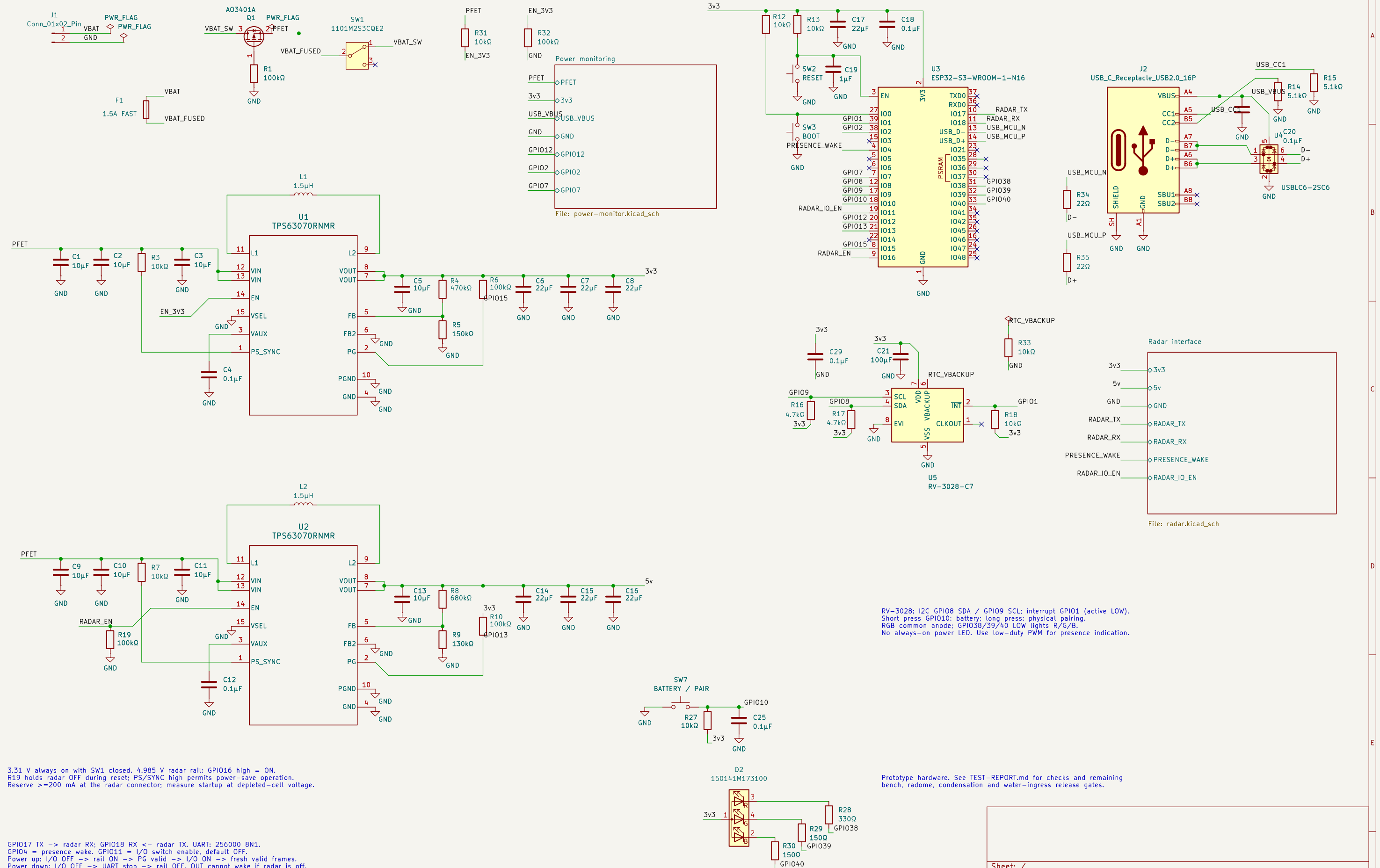


ALEC SENSOR S1 – prototype; bench and ingress tests required.
 3 AA primary cells; no charging. USB-C is data-only; batteries + SW1 ON required.
 SW1: hard battery disconnect, 6 A / 28 VDC. F1: 1.5 A fast fuse; validate inrush and fault clearing.
 RTC time survives ESP deep sleep, but is lost on SW1 OFF or battery removal. Resync before arming.



3.31 V always on with SW1 closed. 4.985 V radar rail: GPIO16 high = ON.
R19 holds radar OFF during reset; PS/SYNC high permits power-save operation.
Reserve ≥ 200 mA at the radar connector; measure startup at depleted-cell voltage.

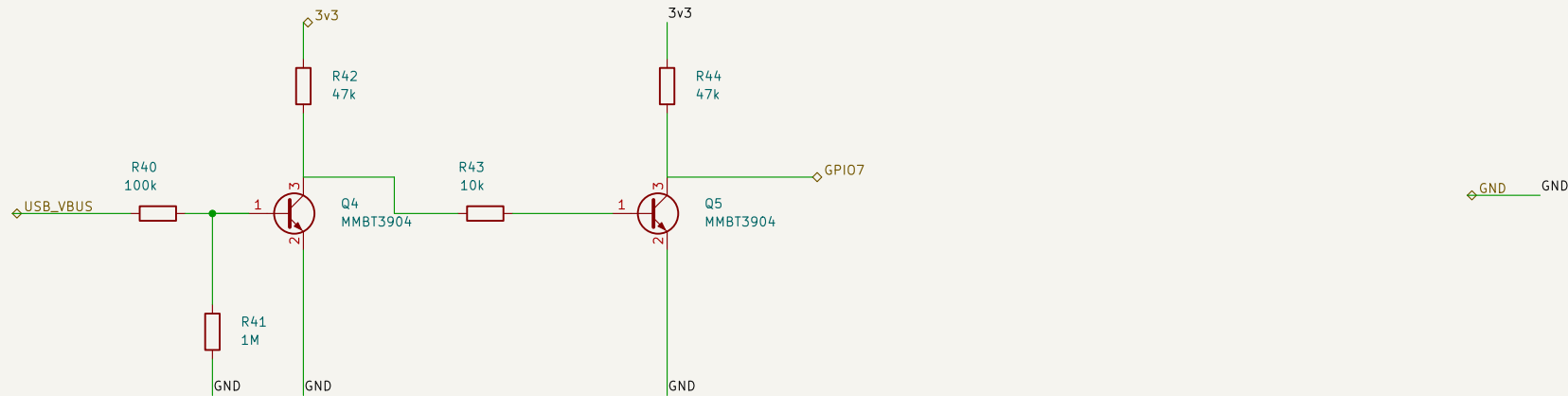
```
GPIO17 TX -> radar RX; GPIO18 RX <- radar TX. UART: 256000 8N1.
GPIO4 = presence wake. GPIO11 = I/O switch enable, default OFF.
Power up: I/O OFF -> rail ON -> PG valid -> I/O ON -> fresh valid frames.
Power down: I/O OFF -> UART stop -> rail OFF. OUT cannot wake if radar is off.
```

Prototype hardware. See TEST-REPORT.md for checks and remaining bench, radome, condensation and water-ingress release gates.

Sheet: /		
File: alec-sensor.kicad_sch		
Title: ALEC Sensor – prototype S1		
Size: A3	Date: 2026-09-07	Rev: S1
KiCad E.D.A. 10.0.1		Id: 1/3

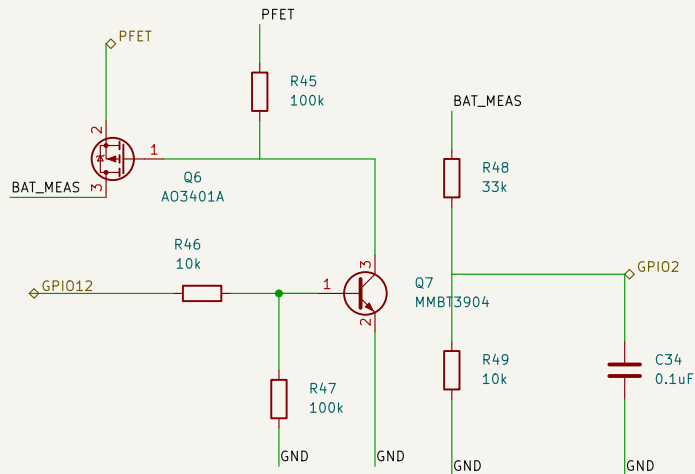
USB VBUS presence – HIGH only with a host supply connected

Two NPN stages keep VBUS away from the GPIO and preserve positive logic.
GPIO7: TinyUSB self_powered=true, vbus_monitor_io=7.
This is cable presence detection, not a precision USB undervoltage comparator.



Switched battery measurement – protected battery node to ADC1 / GPIO2

GPIO12 HIGH enables measurement; wait 10 ms, read calibrated ADC, then drive LOW.
Scale = 4.3; 33k / 10k, 0.1%; 100 nF filter, time constant 0.77 ms.
Designed measurement range 0-6 V; battery chemistry and percentage thresholds remain firmware settings.



Sheet: /Power monitoring/ File: power-monitor.kicad_sch		
Title: ALEC Sensor – power monitoring		
Size: A4	Date: 2026-09-07	Rev: S1
KiCad E.D.A. 10.0.1	Id: 2/3	

LD2410C: regulated 5 V; logic is 3.3 V. J3 numbering follows the module drawing.
This is NOT the pin order of a generic JST cable. Verify assembled socket orientation.
U10 opens all three signal paths when GPIO11 is low; no pull-ups on the unpowered radar side.

Legend:

- 3v3
- 5v
- GND
- RADAR_TX
- RADAR_RX
- PRESENCE_WAKE
- RADAR_IO_EN

U10: TMUX1511PWR

Pin	Label	Signal
14	VDD	3v3
7	GND	GND
S1	RADAR_TX	D1
S2	RADAR_RX	D2
S3	PRESENCE_WAKE	D3
S4	GND	D4
SEL1	RADAR_IO_EN	11
SEL2	RADAR_IO_EN	13
SEL3	RADAR_IO_EN	GND
SEL4	GND	GND

J3: LD2410C - INTERNAL

Pin	Label	Signal
1	LD_TX	LD_TX
2	LD_RX	LD_RX
3	LD_OUT	LD_OUT
4	GND	GND
5	5v	5v

Resistors:

- R54: 100k, connected to RADAR_IO_EN and GND.
- R55: 100k, connected to PRESENCE_WAKE and GND.
- R56: 100k, connected to RADAR_RX and GND.

Capacitors:

- C35: 10uF, connected to 5v and GND.
- C36: 0.1uF, connected to 5v and GND.
- C37: 0.1uF, connected to 3v3 and GND.

Notes:

- ON: rail PG valid before closing U10; ignore startup frames until qualified.
- OFF: open U10 before disabling 5 V. Check reset, brownout and sleep-hold behavior on the bench.
- Powered-radar wake mode: retain 5 V and U10 ON, use OUT only as a wake hint, then verify UART.
- U10 is signal isolation, not a voltage translator. The fourth channel is grounded and disabled.

Sheet Information:

Sheet: /Radar interface/ File: radar.kicad_sch		
Title: ALEC Sensor – switched radar interface		
Size: A4	Date: 2026-09-07	Rev: S1
KiCad E.D.A. 10.0.1	Id: 3/3	

LD2410C: regulated 5 V; logic is 3.3 V. J3 numbering follows the module drawing.
 This is NOT the pin order of a generic JST cable. Verify assembled socket orientation.
 U10 opens all three signal paths when GPIO11 is low; no pull-ups on the unpowered radar side.
 U10 is signal isolation, not a voltage translator. The fourth channel is grounded and disabled.

ON: rail PG valid before closing U10; ignore startup frames until qualified.
 OFF: open U10 before disabling 5 V. Check reset, brownout and sleep-hold behavior on the bench.
 Powered-radar wake mode: retain 5 V and U10 ON, use OUT only as a wake hint, then verify UART.

Sheet: /Radar interface/
 File: radar.kicad_sch

Title: ALEC Sensor – switched radar interface

Size: A4	Date: 2026-09-07	Rev: S1
KiCad E.D.A. 10.0.1		Id: 3/3

Title: ALEC Sensor – switched radar interface		
Size: A4	Date: 2026-09-07	Rev: S1
KiCad E.D.A. 10.0.1		Id: 3/3

Size: A4	Date: 2026-09-07	Rev: S1
KiCad E.D.A. 10.0.1		Id: 3/3

KiCad E.D.A. 10.0.1		Id: 3/3
4	5	6

4	5	6
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